WORKED SOLUTION

STEP II 1997, Question 10

Pure • Collision

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q10

The key idea

Split every velocity into a part along the axis and a part across it. The cylinder is smooth, so it can only push radially: it never changes the axial velocities, and an elastic bounce simply reverses the transverse velocity. Momentum conservation in the explosion then forces the two fragments to fly apart along a single diameter, and the whole problem becomes one-dimensional motion between two walls, with the axial motion going along for the ride.

Solution

Setting up

Model the fragments as particles (they are small), the cylinder as smooth and fixed, and ignore gravity as instructed. Take the z-axis along the cylinder's axis, and let the explosion happen at the origin at time $t = 0$.

Just after the explosion, write the velocity of the fragment of mass qM as an axial part w_1 plus a transverse part $\mathbf{u}_1$ (perpendicular to the axis), and similarly w_2, $\mathbf{u}_2$ for the fragment of mass $(1-q)M$. No external force acts during the explosion, so momentum is conserved:

$$qMw_1 + (1-q)Mw_2 = Mv, \qquad qM\mathbf{u}_1 + (1-q)M\mathbf{u}_2 = \mathbf{0}.$$

The second equation says the transverse velocities point in opposite directions, so both fragments move out from the axis along the same diameter, in opposite senses. Call that diameter the x-direction, with fragment 1 (mass qM) moving towards $x = +L$ at speed u_1 and fragment 2 towards $x = -L$ at speed u_2, where

$$qu_1 = (1-q)u_2. \tag{1}$$

Since $q \leq \frac{1}{2}$, we have $u_1 \geq u_2$: the lighter fragment is the faster.

Each fragment hits the wall head-on (it is moving along a radius), so the smooth wall's impulse is along that radius. That reverses the transverse velocity (the bounce is

perfectly elastic) and leaves the axial velocity alone. So each fragment keeps moving along the same diameter, and its axial velocity is constant throughout.

Showing that $q = \frac{3}{8}$

Axial motion. The axial positions are $z_1 = w_1 t$ and $z_2 = w_2 t$. The fragments meet at some time $t > 0$, so $w_1 = w_2$, and then the momentum equation gives $w_1 = w_2 = v$. Both fragments keep pace with where the original body would have been.

Transverse motion. Fragment 1 reaches $x = L$ at $t = L/u_1$ and then comes back, so after its bounce

$$x_1 = L - u_1\left(t - \frac{L}{u_1}\right) = 2L - u_1 t.$$

In the same way, after its bounce fragment 2 is at $x_2 = -2L + u_2 t$. They collide after both have bounced, so $x_1 = x_2$ gives

$$t = \frac{4L}{u_1 + u_2}, \qquad x = 2L - \frac{4Lu_1}{u_1 + u_2} = 2L\,\frac{u_2 - u_1}{u_1 + u_2}.$$

From (1), $u_2 = \frac{q}{1-q}u_1$, so $u_1 + u_2 = \frac{u_1}{1-q}$ and $u_2 - u_1 = \frac{(2q-1)u_1}{1-q}$. Hence the collision point is

$$x = 2L(2q - 1).$$

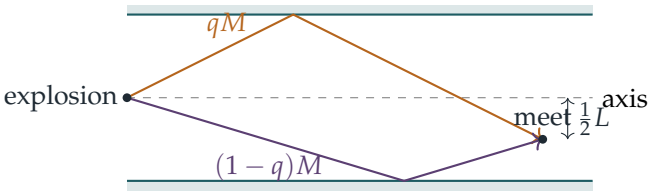
We need $|x| = \frac{1}{2}L$, so $2q - 1 = \pm\frac{1}{4}$, giving $q = \frac{5}{8}$ or $q = \frac{3}{8}$. Since $q \le \frac{1}{2}$,

$$q = \tfrac{3}{8},$$

as required. The collision is at $x = -\frac{1}{2}L$, on the heavier fragment's side of the axis.

Ask yourself Is this really the first meeting, after exactly one bounce each?

With $q = \frac{3}{8}$, (1) gives $u_2 = \frac{3}{5}u_1$. Fragment 1 bounces at $t = L/u_1$ and fragment 2 at $t = \frac{5}{3}L/u_1$, while the collision time is $4L/(u_1 + u_2) = \frac{5}{2}L/u_1$, which is later than both. Could they meet earlier? Before fragment 1 bounces they are moving apart, and on its way back fragment 1 would only reach fragment 2 at $t = 2L/(u_1 - u_2) = 5L/u_1$, long after fragment 2 has turned round. So the meeting we found is the first one, and each fragment has bounced exactly once.



The figure shows the plane containing the axis and the common diameter (axial direction across the page); it is drawn to scale for $q = \frac{3}{8}$ and the speeds found below.

The energy imparted

The collision time is

$$t = \frac{4L}{u_1 + u_2} = \frac{4L(1-q)}{u_1} = \frac{5L}{2u_1}.$$

Setting this equal to $5L/v$ gives $u_1 = \frac{1}{2}v$, and then $u_2 = \frac{3}{5}u_1 = \frac{3}{10}v$.

The axial velocities are unchanged at v, so the kinetic energy gained in the explosion is purely the transverse kinetic energy:

$$\begin{aligned}\Delta E &= \tfrac{1}{2}\cdot\tfrac{3}{8}M(\tfrac{1}{2}v)^2 + \tfrac{1}{2}\cdot\tfrac{5}{8}M(\tfrac{3}{10}v)^2 \\ &= \tfrac{1}{2}Mv^2\left(\tfrac{3}{32} + \tfrac{9}{160}\right) = \tfrac{1}{2}Mv^2\cdot\tfrac{3}{20} = \frac{3Mv^2}{40}.\end{aligned}$$

The velocity after coalescence

Just before the collision, each fragment's transverse velocity has been reversed by its bounce: fragment 1 has transverse momentum $-qMu_1$ and fragment 2 has $+(1-q)Mu_2$. By (1) these add to zero. The axial momentum is $qMv + (1-q)Mv = Mv$. Momentum is conserved in the coalescence, so the combined body of mass M has velocity

$$v \text{ along the direction of the axis,}$$

the same as the original body, but travelling on a line parallel to the axis at distance $\frac{1}{2}L$ from it. All $\frac{3}{40}Mv^2$ of energy from the explosion is lost in the coalescence.

Checks

- $q = \frac{1}{2}$ gives $x = 0$: equal fragments are mirror images and must meet on the axis, as they should.
- With $L = v = 1$, $u_1 = 0.5$ and $u_2 = 0.3$: fragment 1 is at $2 - 0.5\times 5 = -0.5$ and fragment 2 at $-2 + 0.3\times 5 = -0.5$ when $t = 5$. A time-stepped simulation with reflections at $x = \pm 1$ reproduced the meeting at $t = 5.000$, $x = -0.500$, after one bounce each.
- The final transverse momentum $-\frac{3}{8}\cdot\frac{1}{2} + \frac{5}{8}\cdot\frac{3}{10} = 0$, so the final velocity is purely axial.

Common slips

- Letting the wall change the axial velocity. A smooth wall exerts only a normal (radial) force.
- Forgetting that the bounce reverses the transverse momentum. The total transverse momentum only comes back to zero after *both* fragments have bounced.
- Keeping $q = \frac{5}{8}$: that is the same configuration with the fragments' labels swapped, ruled out by $q \le \frac{1}{2}$.

Beyond STEP

Separating motion into components along and across a symmetry direction is the beginning of a big idea: a symmetry of the system gives a conserved quantity. Here the cylinder looks the same all along its length, so axial momentum is conserved even with the walls present (Noether's theorem makes this precise). The question also shows that an explosion followed by a perfectly inelastic collision can give the energy back as heat while the axial motion carries on regardless. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops momentum and centre-of-mass motion in this style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.